Symmetry Reduction in the Detailed Balance Checker

Why Geometric Symmetry Gives a $|G|$-fold Speedup

1 What the checker currently does

The checker verifies *detailed balance* (DB) for every pair of states (s_i, s_j) in a connected component:

$$T(s_i \to s_j)\, \pi(s_i) \;=\; T(s_j \to s_i)\, \pi(s_j), \qquad \pi(s) \propto e^{-\beta E(s)}.$$

For a component with N states this requires evaluating $\binom{N}{2} = N(N-1)/2$ symbolic expressions. For the VMMC `vmmc_lattice.wl` example at NGrid = 3 this is roughly $504 \times 503/2 \approx 126{,}000$ pairs (most trivially zero due to sparse connectivity, but the symbolic work is dominated by the $\sim$thousands of non-trivial ones). The current runtime is around three minutes.

2 The symmetry group of the algorithm

Definition 1 (G-invariance of the transition kernel)**.** Let G be a group acting on the state space $\mathcal{S}$. The transition kernel T is *G-invariant* if

$$T(g \cdot s \to g \cdot s') \;=\; T(s \to s') \quad \forall\, g \in G,\ s, s' \in \mathcal{S}.$$

The energy is G-invariant if $E(g \cdot s) = E(s)$ for all g, s, which implies $\pi(g \cdot s) = \pi(s)$.

For algorithms on a periodic $L \times L$ square lattice, two natural symmetry groups arise.

- **Translations.** $G = \mathbb{Z}_L \times \mathbb{Z}_L$, $|G| = L^2$. The group element (a, b) shifts every site $(r, c) \mapsto (r + a, c + b) \bmod L$. A VMMC or Kawasaki algorithm is translation-invariant whenever the energy depends only on pairwise distances (not absolute positions) and the proposal distribution is spatially uniform. Both conditions hold for `vmmc_lattice.wl`.
- **Full square symmetry (D4 $\times$ translations).** If the algorithm is also invariant under the 8 rotations and reflections of the square, $|G| = 8L^2$.

For concreteness we focus on translations ($|G| = L^2$); every argument applies unchanged to larger groups.

3 Why G-invariance reduces the number of pairs to check

Theorem 1 (Pair-orbit reduction)**.** *Suppose T and E are both G-invariant and G acts freely on $\mathcal{S}$ (no non-identity element fixes any state). Then DB holds for all $N(N-1)/2$ pairs if and only if it holds for a set of representative pairs — one from each G-orbit of unordered pairs. The number of such orbits is*

$$\frac{N(N-1)}{2\,|G|}.$$

Proof. The G-orbit of an unordered pair $\{s, s'\}$ is the set $\{\{g \cdot s,\ g \cdot s'\} : g \in G\}$. Under a free action, all $|G|$ images are distinct, so each orbit has exactly $|G|$ elements. Since every pair belongs to exactly one orbit, the total number of orbits is $\binom{N}{2}/|G| = N(N-1)/(2|G|)$.

Sufficiency. Suppose DB holds for a representative pair $\{s_1, s_2\}$. For any $g \in G$:

$$\begin{aligned} T(g \cdot s_1 \to g \cdot s_2)\, \pi(g \cdot s_1) &= T(s_1 \to s_2)\, \pi(s_1) \\ &= T(s_2 \to s_1)\, \pi(s_2) \\ &= T(g \cdot s_2 \to g \cdot s_1)\, \pi(g \cdot s_2), \end{aligned}$$

where the outer equalities use G-invariance of T and π. Hence DB holds for every pair in the orbit.

Necessity. DB for the orbit member $(g \cdot s_1, g \cdot s_2)$ immediately implies DB for (s_1, s_2) by the same argument with g replaced by g^{-1}. □ □

Corollary 1 (Speedup factor)**.** *With G-invariance the checker needs to evaluate exactly* $\mathbf{1/|G|}$ *as many symbolic DB expressions. The speedup is* $|G|$.

4 Quantitative impact

For NGrid = 3 with translational symmetry

Quantity	Without symmetry	With translations ($|G| = 9$)
States in component	504	504
DB pairs to check	126,756	14,084
Unique symbolic expressions	∼3,000	∼333
Estimated runtime	3 min	∼20 sec

With full D4 + translations ($|G| = 72$)

For a 3×3 torus the full symmetry group (8 rotations/reflections $\times$ 9 translations) has order 72. The same 504-state component would require only $126{,}756/72 \approx 1{,}760$ pair checks and an estimated runtime of ~ 2–3 seconds.

For NGrid = 6 ($N \approx 100{,}000$ states) the untreated checker would require $\sim 5 \times 10^9$ pairs — computationally infeasible — whereas with $|G| = 36$ (translations) this reduces to $\sim 1.4 \times 10^8$ pairs, and with $|G| = 288$ (D4 + translations) to $\sim 1.7 \times 10^7$.

5 The user's concern: what about transitions *within* an orbit?

The key objection is: two states in the *same* G-orbit can transition to each other (e.g. a particle cluster translating to a symmetry-equivalent configuration). Do we not need to check those pairs independently?

The answer is that such pairs are *included* in the orbit-reduction count — they are not a special exception.

Consider two states in the same orbit: $s_1 = c$ (the canonical representative) and $s_2 = h \cdot c$ for some $h \neq e$. This is still a valid unordered pair $\{c, h \cdot c\}$, and its G-orbit is $\{\{g \cdot c,\ gh \cdot c\} : g \in G\}$, which has $|G|$ elements (under a free action). DB for this pair reads

$$T(c \to h \cdot c)\, \pi(c) = T(h \cdot c \to c)\, \pi(h \cdot c).$$

By G-invariance of π this simplifies to

$$T(c \to h \cdot c) \ = \ T(h \cdot c \to c) \ = \ T(c \to h^{-1} \cdot c), \tag{1}$$

where the second equality uses G-invariance of T (apply h^{-1} to both states). Equation (**??**) is a *proposal-symmetry* condition: the probability of moving the cluster by group element h must equal the probability of the inverse move h^{-1}.

For any algorithm whose displacement proposal satisfies $P(\mathbf{d}) = P(-\mathbf{d})$ — which is true by construction for `vmmc_lattice.wl` (the uniform-box list contains $\mathbf{d}$ and $-\mathbf{d}$ as a pair) — equation (**??**) holds automatically. These intra-orbit DB conditions therefore require *no additional checking*: they are guaranteed by the same proposal-symmetry argument that underlies the VMMC detailed-balance proof.

Even without that argument: intra-orbit pairs number at most $M \cdot |G|(|G|-1)/2 \approx N|G|/2 = O(N)$, whereas inter-orbit pairs number $M(M-1)|G|^2/2 \approx N^2/(2|G|) = O(N^2)$. For large N the intra-orbit contribution is negligible, and the dominant $O(N^2)$ cost still shrinks by exactly $|G|$.

6 What implementation is needed

Three components are required to realise the speedup in practice.

1. **Verify G-invariance.** For each canonical seed c, compare the BFS output (transition probabilities from c) with the BFS output from a non-canonical orbit member $g \cdot c$. G-invariance holds iff the two outputs agree after applying g^{-1} to all successor states. This is an exact symbolic check with negligible overhead.

2. **Canonicalise states.** Map each state to the lexicographically smallest member of its G-orbit before adding it to the "already seen" set. Only one BFS is run per orbit; BFS from $g \cdot c$ is replaced by looking up the result for c and applying g.

3. **Check one pair per G-orbit.** For each pair of canonical representatives (c_i, c_j), check only the pair (c_i, c_j) itself (or any single orbit representative). The $|G| - 1$ other pairs in the same orbit are certified by Theorem 1 without additional computation.

None of these steps changes the *algebraic* content of the DB check; they reduce the *number of times* it is performed.